K L Deemed to be University

Department of computer science and Engineering -1 -- KLBCH
Course Handout
2026-2027, Even Sem

Course Title	**: ADVANCED OBJECT ORIENTED PROGRAMMING**
Course Code	**: 25CS2205E**
L-T-P-S Structure	**: 2-0-2-4**
Pre-requisite	: 25SC1105E-PROBLEM SOLVING THROUGH PROGRAMMING (JAVA)
Credits	**: 4**
Course Coordinator	**: Karumuri Sri Rama Murthy**
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Teaching Associates	:
Syllabus :	M1 Modelling I — Objects that hold their shape, and types that carry the design (Weeks 01, 05) Maps to CO1 What an object actually is: the guarantee it makes; writing invariants as sentences before any field; single-field, multi-field and temporal invariants; the reachable state space and the sentence test. Where guarantees break: the five doorways — public setters, escaping references, mutable elements inside immutable collections, constructors that validate nothing, and equality that lies; tell, don't ask. Money is not a number: paise as long, currency-aware, immutable value with plus/minus/times/allocate; why a double can never hold a fare (binary expansion and accumulation). Identity, equality and the hash contract: entity versus value by three questions; equals/hashCode as a contract; safe cache keys. Construction, escape and defensive copying: construction as the gatekeeper; aliasing doorways; immutability and thread-safety for free. Making illegal states unrepresentable: typed ids (TicketId, SpotId, VehicleId, BookingId), sealed hierarchies, and a list of expressions that must fail to compile as the real test suite. Generics, bounds and what erasure costs; variance derived in four lines and PECS stated as a consequence; collections chosen by workload — read share, size, ordering, concurrency. Language annex topics taught here: nested and anonymous classes (static nested by default, the inner-class leak), varargs, boxing and the unboxing NullPointerException (Annex C.4); the same ideas in C++ and Python — RAII, value semantics, runtime permissiveness (Annex C.5). Design autopsy, invariant-breaker instrument, debugging lab, and the Waypoint core-model design challenge. M2 Modelling II — Contracts before code (Week 02, with Week 04's Liskov and interface segregation) Maps to CO2 Design by contract: for any method, its precondition, postcondition and invariant stated before the body; the six clauses that are part of a contract but not of a signature. Liskov, stated precisely: the direction rule for pre- and postconditions; substitutability judged from the caller's side; the four

	violation signals — throwing override, empty override, added precondition, instanceof. Interface, abstract class, or sealed: choosing by the cost of each in one sentence; sealed types as closed contracts. Role interfaces at real seams; interface segregation argued as a testing principle; contract tests that judge every implementation with one shared suite, plus a saboteur that must be caught. Writing SpotAllocator's contract first — in Javadoc, before any implementation — then three implementations judged by one suite. Contract violations that produce real JDK error messages: which clause was broken, and by whom — the caller or the implementer. Design autopsy, design lab, debugging lab, and the allocation-contract design challenge. M3 The cost of change — coupling, SOLID, and patterns costed (Weeks 03, 04, 07, 08, 09) Maps to CO3 Blast radius: design you can measure — predicting a change's cost in four numbers before writing code, then checking it; coupling priced in seven flavours; cohesion computed from the method/field matrix. Dependency inversion and the composition root; the delete-the-infrastructure test; ports in the domain, adapters in infrastructure depending inward; the Law of Demeter. SOLID applied, not recited: single responsibility as one actor; open–closed as choosing an axis from commit history; Liskov and interface segregation as the two about callers; when to apply none of them; inheritance versus composition class counts for two-axis variation. Creational patterns and when they hurt: what construction decides — timing, family, rules, cost; Factory Method and Abstract Factory; Builder for rules not parameters; Singleton's four costs, Prototype, Object Pool and dependency injection. Structural patterns — composing without inheriting: Adapter as defending your vocabulary; Decorator and why layer order is observable; Proxy versus Facade; Composite, Flyweight, Bridge and the tree assumption. Behavioural patterns and honest state machines: flags as an unnamed state machine — combination counts, legal states, total transition tables; Strategy, Template Method or a sealed switch by binding time and axis count; Observer done safely and the four decisions hidden in "publish an event"; Command, Chain of Responsibility, Visitor, Mediator. Anti-patterns and restraint: God class, anaemic model, pattern-itis; YAGNI; refactoring toward patterns with characterisation tests green at every commit. Lambdas, streams and functional interfaces where patterns need them: strategies as lambdas, the standard functional interfaces, stream laziness and single use, and why parallelStream is a concurrency decision in disguise (Week 09, Annex C.1). Reflection and annotations for checking code rather than doing things at runtime: an annotation plus a test turning a design rule into a build failure (Annex C.3). Design challenges: invert Waypoint; the god-class teardown in five commits; lifetimes, families and stages; the gateway stack and the facility tree; machines, events and a lift bank. M4 Failure as a first-class design concern (Week 06, with Week 11's outbox and compensation) Maps to CO4 Three species of failure — expected outcome, programmer error, transient infrastructure — as a sealed failure taxonomy; checked versus unchecked as a design decision; Result types where control flow is normal. A timeout is an unknown outcome, not a failure: the third state it requires; what the caller knows about whether the charge occurred. Idempotency, the load-bearing clause: four techniques to make a non-idempotent operation idempotent, and how to choose between them. Timeouts, retries and the storm they cause: how retries turn a degraded dependency into a total outage; backoff, jitter and budgets; circuit

	breakers and the three fixes in order. Compensation: undoing what cannot be undone; sagas; the dual write and the outbox; the guarantee an outbox gives and the idempotent consumer it requires. The chaos test: a 30% injected failure rate across three modes that must produce zero double charges without slowing the happy path. Design autopsy, incidents that produce no stack trace, and the make-payments-survivable design challenge. M5 Runtime — Concurrency as design; persistence, aggregates and transaction boundaries (Weeks 10, 11) Maps to CO5 The three safe states: every mutable field is immutable, confined, or guarded by X — and every access must respect its category; a scan test that lists unclassified fields. Check-then-act in all its disguises, including its database form; fixing it by making the operation atomic; optimistic concurrency and versioning. Contention regimes chosen from a measured collision rate; locks, atomics, executors, CompletableFuture; deadlock ordering; queues as back-pressure. The Java memory model: happens-before, volatile, unsafe publication and the four ways to publish safely; correctness across two JVMs, where the design changes. Aggregates are transaction boundaries: drawing them with four questions; local invariants that imply the global one; scaling consequences and hotspots. Repository as a collection illusion and Unit of Work; isolation by anomaly — the four anomalies, which levels prevent each, write skew as check-then-act; fixing without Serializable. ORM impedance, the dual write, and an outbox that survives a crash mid-batch. Files, streams and serialization as an adapter: serialized form as a published contract, schema evolution rules, try-with-resources, and why Java serialization is a cautionary tale (Week 11, Annex C.2). Design challenges: correct under load and across processes; boundaries, outbox and write skew. M6 Architecture at scale, and shipping it — the capstone (Week 12) Maps to CO6 Bounded contexts: where the vocabulary changes; anti-corruption layers; the modular monolith and its numeric extraction triggers. Numbers first, then boxes: capacity estimation for Waypoint at 50× load; caching, sharding and the arithmetic that justifies each box. Consistency per operation, degraded modes, and observability — structured logs, health, and metrics on the four decisions that matter. Module boundaries enforced by architecture tests; booking, pricing, billing and notification modules each owning its schema. The HLD dossier — six artefacts; twelve ADRs, each naming a rejected alternative and a cost; the ten-minute recorded walkthrough and the evidence pack. UML as a graded deliverable: class, sequence and state diagrams with worked Waypoint examples, one diagram specified per week (Annex D). Engineering practice: CI as a design instrument, security by design, data protection, cost as an axis, postmortems and SLOs, and design judgement when a model writes the first draft (Annex E). The mock loop: LLD and HLD interview rounds run on the student’s own repository; retrieval across the whole course.
Text Books :	1. Effective Java Joshua Bloch Addison-Wesley 2018 (3rd Ed.) 2. A Philosophy of Software Design John Ousterhout Yaknyam Press 2021 (2nd Ed.) 3. Domain-Driven Design: Tackling Complexity in the Heart of Software Eric Evans Addison-Wesley 2003
Reference Books :	1. Implementing Domain-Driven Design Vaughn Vernon Addison-Wesley 2013 2. Design Patterns: Elements of Reusable Object-Oriented Software Erich Gamma, Richard Helm, Ralph Johnson & John Vlissides Addison-Wesley 1994 3. Refactoring: Improving the Design of Existing Code Martin

	Fowler Addison-Wesley 2018 (2nd Ed.)
MOOCS :	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153
Course Rationale :	Advanced Object-Oriented Programming is the Regular-track sister of PSPJ's object model and the PBL design studio for CSE: a twelve-week, project-based course in software design — LLD and HLD — rather than in syntax students already have. "Design is what you can change on a Tuesday": the course teaches the decisions that determine whether next quarter's requirement costs two files or two weeks, and the habit of pricing every decision aloud. INSIDE-OUT, in five movements: Modelling (objects that hold their shape, contracts before code, the cost of change), Principles (SOLID applied, the type system as a design tool, failure as a first-class design concern), Patterns (creational, structural and behavioural, each selected by problem shape and costed), Runtime (concurrency as design, persistence and aggregates), and Architecture & Capstone (bounded modules, capacity arithmetic, shipping it). Every week is a full build on one repository — Waypoint, an urban mobility and payments platform — with four bespoke interactive instruments, a debugging lab of four real failures, starter code with failing tests, and a gated reference; six studio builds (SeatLock, LedgerPay, FleetMatch, GateKeeper, HeraldQ, MatchBook) are the transfer test, and each is a famous interview problem in domain clothes. Language topics — generics, lambdas and streams, reflection, serialization, JVM behaviour, C++ and Python equivalents — are taught where a design problem needs them and collected in the Language Annex, not as chapters. Honest scope: it does not teach DSA, a framework, front-end, DevOps tooling or behavioural rounds. Pre-requisite: PSPJ — Java 17+, classes and interfaces, generics as a consumer, collections, exceptions; plus Gradle/Maven, JUnit 5, Git, and basic SQL from Week 11; concurrency is not assumed. Enables: SDSR — System Design for Scalability, and every backend, platform and LLD/HLD interview-loop course downstream. Placement corridor: the claim the course makes is precise — a student who completes it can design, defend and extend a non-trivial system and perform in the LLD and HLD rounds of a product-company interview loop; ₹8–22 LPA backend and platform-engineering roles at India product companies, with the DSA leg covered by the DSA plan and the behavioural leg by Professional Presence.
Course Objectives :	To equip students with the knowledge and practical skills to analyse, design, develop, evaluate, and extend maintainable, reliable, reusable, and scalable software systems by applying advanced object-oriented modelling, design principles, design patterns, type-driven programming, concurrency, persistence, and low-level and high-level software architecture techniques through project-based learning, while enabling them to make informed design decisions, analyse trade-offs, debug real-world failures, and develop industry-ready software solutions.

COURSE OUTCOMES (COs):

CO	Course Outcome (CO)	PO/PSO	Blooms

NO			Taxonomy Level (BTL)
CO1	Model a domain as objects that enforce their own invariants, distinguishing entities from values and eliminating illegal states.	PO2,PSO1,PO1	3
CO2	Specify and verify behavioural contracts — preconditions, postconditions, invariants — and detect substitutability violations.	PO1,PO2,PSO1	4
CO3	Measure and reduce the blast radius of change, and justify each abstraction by a named axis and a stated cost.	PO2,PO3,PO10,PSO1	5
CO4	Design systems that survive failure — classifying failures, applying idempotency, retries, breakers and compensation.	PSO1,PSO2,PO4,PO3,PO2,PO1	6
CO5	Reason about correctness under concurrency and across processes, choosing a contention regime from evidence.	PO1,PO2,PO4,PSO1	4
CO6	Compose a system from bounded modules, derive its architecture from capacity estimates, and defend it under questioning.	PO2,PO3,PO4,PO5,PO9,PO10,PO11,PSO1,PSO2	6

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5	COI-6
CO1	3	**Btl-2** Understanding domain concepts as classes that strictly encapsulate state and enforce their own internal invariants.	**Btl-3** Apply Design object constructors and mutators to completely eliminate illegal or invalid object states.			
CO2	4	**Btl-2** Understanding behavioral contracts programmatically using assertions, defensive checks, and design-by-	**Btl-3** Applying inheritance and polymorphism hierarchies to detect and prevent Liskov Substitution	**Btl-4** Analyzing contract enforcement mechanisms to ensure robust component interaction and		

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5	COI-6
		contract principles.	Principle (LSP) violations.	reliable error handling.		
CO3	5	**Btl-2** Understanding software abstractions against specific, named axes of change (e.g., extensibility, maintainability).	**Btl-3** Applying the introduction of design patterns or architectural abstractions by weighing their benefits against stated implementation costs.	**Btl-4** Analyzing coupling and cohesion metrics to minimize cascading failures during system evolution.	**Btl-5** Evaluate refactoring strategies to ensure long-term architectural stability and maintainability.	
CO4	6	**Btl-2** Understanding Idempotent operations to ensure safe operation retries without unintended side effects.	**Btl-3** Applying retry policies and exponential backoff strategies for transient fault handling.	**Btl-4** Analyzing circuit breaker patterns to prevent cascading failures across service boundaries.	**Btl-5** Evaluate compensation mechanisms and rollback transactions for failing distributed workflows.	**Btl-6** Create resilient system architectures capable of graceful degradation under partial failures.
CO5	4	**Btl-2** Understanding inter-process communication and synchronization bottlenecks in concurrent systems.	**Btl-3** Applying evidence and performance metrics to select the appropriate contention regime (e.g., optimistic vs. pessimistic locking).	**Btl-4** Analyzing concurrency hazards such as deadlocks, race conditions, and livelocks within object-oriented code.		
CO6	6	**Btl-2** Understanding system architecture and resource boundaries directly from quantified capacity estimates and load requirements.	**Btl-3** Apply cohesive subsystems adhering to Clean Architecture and SOLID design principles.	**Btl-4** Analyze interfaces and communication contracts between architectural tiers.	**Btl-5** Evaluate architectural trade-offs regarding scalability, throughput, and maintenance complexity.	**Btl-6** Create the proposed system architecture and design decisions under technical peer review and questioning.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Object Guarantees, Invariants & State Space	TB1 [Ch.2, Pg 21-40]	Chalk,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
2	CO1	COI-2	The five invariant doorways: public setters, escaping references, mutable collections, liar equality	TB2[ch2, pg15-32]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
3	CO1	COI-3	Money as an Immutable Value: Paise, Currency, Equality, and Identity vs Value	TB2[Ch3, Pg 33-58]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						(Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
4	CO1	COI-3	Defensive Copying, Typed IDs, Illegal States, and Collections Overview	TB2[Ch-4, Pg 65-90]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
5	CO2	COI-3	Design by Contract: Preconditions, Postconditions, and Invariants	TB1[Ch-8,Pg 70-190]	Chalk,LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
6	CO2	COI-3	Liskov Substitution Principle and Caller-Side Substitutability Violations	TB2[Ch-4,Pg 95-115]	Chalk,LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
7	CO2	COI-2	Interfaces, Abstract Classes, Sealed Types, and Role Seams	TB1[Ch-9, Pg 215-240]	Chalk,LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
8	CO3	COI-3	Measuring Blast Radius	TB2[Ch5, Pg 120-145]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate +

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
9	CO3	COI-2	Dependency Inversion adn SOLID	RB1[Ch1, Pg 1-35]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
10	CO3	COI-2	Creational and Structural Patterns costed, Factory Builder, SIngleton, Adapter, Decorator, Proxy	RB1[Ch-3, Pg 80-130]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
11	CO3	COI-2	Behavioural Patterns and State Machines, Strategy, Observer,	RB1[Ch6 Pg 200-260]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
			Event safety, Command, Lambdas/ Streams			Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
12	CO3	COI-3	Anti-Patterns, Build Rules, God class teardown,YAGNI, reflection/annotation compile checks	RB1[Ch11, Pg 380-410]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
13	CO4	COI-3	Sealed Failure Taxonomy: 3 failure species, Checked vs Unchecked, Result Type vs Control Flow	TB2[Ch-10, Pg 280-305]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						Evaluation,Term End Project Defence & Demonstration
14	CO4	COI-3	Timeouts as Unknown Outcomes, Idempotency, and Load-Bearing Clauses	RB2[Ch-6, Pg 250-280]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
15	CO4	COI-3	Retries, Exponential Backoff, Jitter, Budgets, and Circuit Breakers	RB2[Ch-8, Pg 281-315]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
16	CO4	COI-3	Compensation, Sagas, Dual Writes, Outbox Pattern, and Chaos Testing	RB2{Ch11, Pg 390-420]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
17	CO5	COI-2	Safe Concurrency States and Check-Then-Act Bugs	TB1[Ch11, Pg 285-310]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
18	CO5	COI-3	Locks, Atomics, Executors, CompletableFuture, Deadlock, and Backpressure	TB2{Ch11, Pg 320-350]	Chalk,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
19	CO5	COI-2	ava Memory Model, Happens-Before, Volatile, and Safe Publication	TB2[Ch11, Pg 351-380]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
20	CO5	COI-2	Aggregates, Repositories, Unit of Work, Isolation Anomalies, and Write Skew	RB2[Ch7, Pg 220-255]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
21	CO6	COI-3	Bounded Contexts, Anti-Corruption Layers, and Modular Monolith	RB2[Ch-9, Pg 322-355]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate +

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
22	CO6	COI-3	Architecture Tests, HLD Dossier, ADRs, and Evidence Packs	RB2{Ch11, Pg 425-460]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
23	CO6	COI-2	Architecture Tests, HLD Dossier, ADRs, and Evidence Packs	RB2[Ch11, Pg 425-250]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester Exam,Global Challenges - Participation,In-Term Practical Examination,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration
24	CO6	COI-3	UML, Engineering Practice, CI/CD, Security, Cost, SLOs,	RB2[Ch11, Pg 475-520]	LTC,PPT,Talk	Active Learning Engagement (In-Class),End Semester

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
			and Mock Review			Exam,Global Challenges - Participation,Mid-Term Examination (Descriptive),MOOCs Evaluation (Certificate + Test),Practical Continuous Evaluation,Project Continuous Evaluation,Term End Project Defence & Demonstration

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduce AOOP syllabus, PBL framework, Waypoint Smart Parking core model.	3	Talk	Just in-time teaching
20	Define object guarantees, reachable state space, writing invariants as sentences.	3	Chalk	Case Study Based Learning
25	Apply sentence tests to validate domain fields before writing class declarations.	3	LTC	Problem-Based Learning

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Gist of invariants and state space constraints	3	Talk	Quiz/Test Questions
20	explain 5 invariant doorways: public setters, escaping references, liar equality	3	Chalk	Case Study
25	instrument invariant-breakertests to expose mutable collection leaks	3	LTC	Problem-Based Learning

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Review of Aliasing doorways and defensive copies	3	Talk	Just-in-time Teaching
20	Model money as paise, long, currency-aware value objects: binary expansion double flaws	3	Chalk	Problem-Based Learning
25	Implement equals/hashcode value contracts and verify safe cache keys	3	LTC	Lab Taken to Class

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap mutable state and introduce reference aliasing.	3	PPT	Problem-Based Learning
20	Explain defensive copying, typed identifiers, illegal states and Java Collections. Demonstrate how returning a mutable collection can expose internal state.	3	PPT	Problem-Based Learning
25	Students analyze unsafe getters and constructors, identify identifier-mixing risks and suggest defensive-copying and validation solutions.	3	PPT	Problem-Based Learning

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Design by Contract	4	Talk	Problem-Based Learning
20	Preconditions, Postconditions and Invariants	4	Chalk	Problem-Based Learning
25	Preparing a Method Contract Sheet	4	LTC	Problem-Based Learning

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-	Active

			Learning Methods	Learning Methods
5	Introduction to Substitutability	4	Talk	Case Study Based Learning
20	Liskov Substitution Principle	4	PPT	Case Study Based Learning
25	Analyzing Subtype Contract Violations	4	LTC	Case Study Based Learning

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Abstraction	4	Talk	Problem-Based Learning
20	Interfaces, Abstract Classes and Sealed Types	4	PPT	Problem-Based Learning
25	Selecting an Appropriate Abstraction	4	Chalk	Problem-Based Learning

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Contract Testing	5	Talk	Problem-Based Learning
20	Contract Tests and SpotAllocator Design	5	PPT	Problem-Based Learning
25	Designing Tests for SpotAllocator	5	LTC	Problem-Based Learning

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-	Active

			Learning Methods	Learning Methods
5	Introduction to Dependency Inversion	5	Talk	Problem-Based Learning
20	Dependency Inversion and Composition Root	5	PPT	Problem-Based Learning
25	Refactoring for Infrastructure-Independent Tests	5	LTC	Problem-Based Learning

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to SOLID principles; identify problems in a class having multiple responsibilities.	5	Talk	Quiz/Test Questions
20	Explain Single Responsibility, Open–Closed, Liskov Substitution, Interface Segregation and Dependency Inversion principles with simple Java examples.	5	PPT	Problem-Based Learning
25	Explain Factory, Builder, Adapter, Facade and Decorator patterns. Evaluate which pattern is suitable for a given Waypoint design problem.	5	PPT	Case Study Based Learning

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to behavioural patterns and booking status changes.	5	Talk	Quiz/Test Questions
20	Explain Strategy, Observer and State patterns. Introduce state machines and valid state transitions using a Booking example.	5	PPT	Problem-Based Learning
25	Create a booking state-transition model. Identify anti-patterns such as scattered conditions, invalid transitions and hidden states.	5	LTC	Problem-Based Learning

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Anti-Patterns: Explain the meaning of anti-patterns and code smells. Introduce the God Class problem using a simple Java example.	5	Talk	Quiz/Test Questions
20	God Class Teardown, Build Rules and YAGNI: Explain how a God Class handles too many responsibilities. Demonstrate breaking it into smaller classes. Introduce basic build rules and explain YAGNI (You Aren't Gonna Need It) with examples of avoiding unnecessary features.	5	PPT	Case Study Based Learning
25	Reflection, Annotations and Compile-Time Checks: Explain Java reflection and annotations. Discuss the difference between runtime reflection and compile-time validation. Demonstrate how annotation processors, compiler checks and build rules can identify design or coding violations. Conduct a small code-review activity.	5	LTC	Problem-Based Learning

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to failures: invalid booking input, programming error and temporary service failure.	6	Chalk	Quiz/Test Questions
20	Explain expected/domain failures, programmer errors and transient infrastructure failures with simple examples.	6	Talk	Problem-Based Learning
25	Classify Waypoint failure scenarios and prepare a handling table showing when to reject, log, retry or return an error.	6	PPT	Problem-Based Learning

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to timeout: what happens when a booking request receives no response?	6	Talk	Quiz/Test Questions
20	Explain unknown outcomes, idempotency, idempotency keys and the conditions required for safe repeated requests.	6	PPT	Case Study Based Learning
25	Design an idempotent booking request flow. Demonstrate how the system prevents duplicate bookings when the same request is submitted twice.	6	LTC	Problem-Based Learning

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to retry mechanisms and problems caused by repeated requests.	6	Talk	Quiz/Test Questions
20	Explain retry limits, exponential backoff, jitter, retry budgets and circuit-breaker states (Closed, Open and Half-Open).	6	PPT	Problem-Based Learning
25	Prepare a retry policy for a Waypoint service with retry limits, delay intervals and circuit-breaker conditions.	6	LTC	Problem-Based Learning

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to multi-step booking operations and failures during transaction processing.	6	Talk	Quiz/Test Questions
20	Explain compensation, saga pattern, dual-write problem, outbox pattern and chaos testing.	6	PPT	Case Study Based Learning
25	Design a booking and notification workflow. Identify failure points and prepare recovery steps using compensation and the outbox pattern.	6	LTC	Problem-Based Learning

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to concurrent booking requests and shared data.	4	Talk	Quiz/Test Questions
20	Explain immutable state, thread-confined state, guarded state, race conditions and check-then-act problems.	4	PPT	Problem-Based Learning
25	Analyze two threads attempting to reserve the same parking spot. Identify the race condition and suggest a safe solution.	4	LTC	Problem-Based Learning

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to multithreading and the need to coordinate multiple tasks.	4	Talk	Quiz/Test Questions
20	Explain synchronized methods, locks, atomic variables, ExecutorService and CompletableFuture with simple Java examples.	4	PPT	Problem-Based Learning
25	Analyze simple deadlock and thread-overload scenarios. Compare suitable concurrency tools and identify how backpressure helps control overload.	4	LTC	Problem-Based Learning

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to shared variables and visibility problems between threads.	4	Talk	Quiz/Test Questions
20	Explain Java Memory Model, happens-before relationship, volatile keyword and safe publication.	4	PPT	Problem-Based Learning
25	Analyze a Java program in which one thread updates a flag and another thread reads it. Identify visibility problems and suggest a safe solution.	4	LTC	Problem-Based Learning

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to transactions and problems caused by simultaneous updates.	4	Talk	Quiz/Test Questions
20	Explain aggregates, repositories, Unit of Work and transaction boundaries using a simple booking example.	4	PPT	Case Study Based Learning
25	Explain isolation anomalies and write skew. Analyze two concurrent transactions that together violate a business rule and suggest a solution.	4	LTC	Problem-Based Learning

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Waypoint capabilities such as Booking, Pricing and Notification.	6	Talk	Quiz/Test Questions
20	Explain bounded contexts, context maps, anti-corruption layers and modular monolith architecture.	6	PPT	Problem-Based Learning
25	Create a simple Waypoint context map and identify suitable module boundaries between Booking, Pricing and Notification.	6	LTC	Problem-Based Learning

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Introduction to increasing user requests and the need for application scalability.	6	Talk	Quiz/Test Questions
20	Explain basic capacity estimation, caching, sharding, consistency, performance metrics and SLOs.	6	PPT	Case Study Based Learning
25	Estimate a simple Waypoint request load and prepare a scaling plan using caching, sharding and suitable monitoring metrics.	6	LTC	Problem-Based Learning

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ntroduction to architecture decisions and the importance of documenting design choices.	6	Talk	Quiz/Test Questions
20	Explain architecture tests, High-Level Design (HLD) dossier, Architecture Decision Records (ADRs) and evidence packs.	6	PPT	Case Study Based Learning
25	Prepare a sample ADR for a Waypoint design decision and identify evidence required to support the decision.	6	LTC	Problem-Based Learning

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to the final architecture dossier and project review requirements.	6	Talk	Quiz/Test Questions
20	Explain UML diagrams, engineering practices, CI/CD, security considerations, cost estimation and SLOs.	6	PPT	Case Study Based Learning
25	Conduct a mock architecture review. Evaluate UML diagrams, ADRs, testing evidence, security, cost and operational requirements.	6	LTC	Case Study Based Learning

Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Set Up Waypoint Core Model and Write Sentence Invariants	CO1
2	Defensive Copying and Immutable Money	CO1
3	Design by Contract and SpotAllocator	CO2
4	Liskov Substitution and Contract Testing	CO2
5	Dependency Mapping and Dependency Inversion	CO3
6	God Class Teardown and Behavioural Pattern	CO3
7	Failure Handling and Idempotency	CO4
8	Retry Strategy and Recovery Mechanism	CO4
9	Concurrency and Race Condition Correction	CO5
10	Transaction Boundaries and Isolation Anomalies	CO5

Tutorial Session no	Topics	CO-Mapping
11	Architecture Rules and Dependency Testing	CO6
12	Waypoint Architecture Dossier and Defence	CO6

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction to Waypoint project	3	Talk	Think / Pair / Share
25	Eclipse project setup	3	Chalk	Problem-Based Learning
25	Domain objects and fields	3	Chalk	Problem-Based Learning
30	Writing sentence invariants and validation	3	LTC	Problem-Based Learning
10	Testing and submission	3	LTC	Peer Survey

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Inspect unsafe Java code	3	Chalk	Problem-Based Learning
25	Identifying unsafe state access	3	Talk	Problem-Based Learning
25	Defensive copying	3	Chalk	Peer Survey
30	Implement immutable Money	3	Chalk	Problem-Based Learning
10	Test immutable Money	3	LTC	Peer Review

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction to method contracts	4	Talk	Think / Pair / Share
25	Inspect SpotAllocator interface	4	Chalk	Problem-Based Learning
25	Writing contract statements	4	Chalk	Problem-Based Learning
30	Implement contract validation	4	LTC	Problem-Based Learning
10	Contract review	4	LTC	Peer Review

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction to base and subtype classes	4	Talk	Problem-Based Learning
25	Identify contract violations	4	Chalk	Problem-Based Learning
25	Contract test design	4	Chalk	Problem-Based Learning
30	Write and repair contract tests	4	PPT	Problem-Based Learning
10	Execute tests and report findings	4	PPT	Problem-Based Learning

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Inspect class dependencies	5	Talk	Problem-

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				Based Learning
25	Dependency tracing	5	Chalk	Problem-Based Learning
25	Compare coupling and cohesion	5	Chalk	Problem-Based Learning
30	Apply dependency inversion	5	LTC	Peer Review
10	Testing without infrastructure	5	LTC	Peer Review

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Identify God Class	5	Talk	Problem-Based Learning
25	Responsibility identification	5	Chalk	Problem-Based Learning
25	Design smaller classes	5	Chalk	Problem-Based Learning
30	Refactoring and behavioural pattern	5	LTC	Problem-Based Learning
10	Compare before and after design	5	LTC	Peer Review

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Review failure scenarios	6	Talk	Problem-Based Learning
25	Success and failure representation	6	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Duplicate booking identification	6	Chalk	Problem-Based Learning
30	Implement idempotency	6	LTC	Problem-Based Learning
10	Test duplicate and failure cases	6	LTC	Problem-Based Learning

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Timeout and retry requirements	6	Talk	Problem-Based Learning
25	Retry limit and backoff	6	Chalk	Problem-Based Learning
25	Circuit-breaker states	6	Chalk	Problem-Based Learning
30	Outbox and recovery sequence	6	LTC	Problem-Based Learning
10	Record expected behaviour	6	LTC	Problem-Based Learning

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Inspect shared mutable fields	4	Talk	Problem-Based Learning
25	Identify unsafe shared state	4	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Reproduce check-then-act race	4	Chalk	Problem-Based Learning
30	Apply concurrency protection	4	LTC	Problem-Based Learning
10	Test the corrected program	4	LTC	Problem-Based Learning

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Identify related data changes	4	Talk	Problem-Based Learning
25	Aggregate and repository mapping	4	Chalk	Problem-Based Learning
25	Transaction scenario	4	Chalk	Problem-Based Learning
30	Analyze isolation anomalies	4	LTC	Case Study Based Learning
10	Present transaction design	4	LTC	Case Study Based Learning

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Review Waypoint modules	6	Talk	Problem-Based Learning
25	Define dependency rules	6	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Design architecture tests	6	Chalk	Problem-Based Learning
30	Implement or prototype architecture test	6	LTC	Peer Review
10	Execute and record evidence	6	LTC	Peer Review

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Prepare dossier checklist	6	Talk	Problem-Based Learning
25	Review UML and component mapping	6	Chalk	Problem-Based Learning
25	Review architecture decisions	6	Chalk	Problem-Based Learning
30	Present and defend architecture	6	LTC	Problem-Based Learning
10	Feedback and improvement	6	LTC	Problem-Based Learning

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Explore Waypoint Domain Architecture and Draft Invariants	CO1
2	Analyze Single-Field, Multi-Field and Temporal Invariants	CO1
3	Identify Aliasing Doorways and Escaping-Reference Hazards	CO1
4	Design Typed IDs and Make Illegal States Unrepresentable	CO1
5	Specify Preconditions and Postconditions for Domain Services	CO2

Skilling session no	Topics/Experiments	CO-Mapping
6	Analyze Caller-Side Substitutability and Override Rules	CO2
7	Construct Role Interfaces and Shared Contract-Test Suites	CO2
8	Diagnose Java/JDK Errors Caused by Contract Violations	CO2
9	Calculate Cohesion Matrices and Estimate Change Coupling	CO3
10	Apply Delete-the-Infrastructure Test and Law of Demeter	CO3
11	Refactor with Characterization Tests Green at Each Step	CO3
12	Use Lambdas and Annotations for Clarity and Build-Time Rules	CO3
13	Design Sealed Failure Taxonomies and Result Control Flows	CO4
14	Engineer Retries with Exponential Backoff, Jitter and Budgets	CO4
15	Design Compensating Transactions and Sagas	CO4
16	Simulate Chaos Injection and Validate Duplicate-Charge Prevention	CO4
17	Audit Mutable-Field Access and Classify Confinement/Guarding Strategies	CO5
18	Manage Contention Using Atomics, Locks and CompletableFuture	CO5
19	Analyze Happens-Before Semantics Across Threads	CO5
20	Analyze Isolation Anomalies and Write Skew	CO5
21	Define Anti-Corruption Layers for Bounded-Context Integration	CO6
22	Estimate Capacity for 50× Load and Discuss Caching/Sharding Trade-offs	CO6
23	Formulate ADRs with Alternatives, Trade-offs and Rejected Options	CO6
24	Prepare HLD/UML Evidence Pack and Mock Defence	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce the Waypoint application. Explain Booking, Spot, Vehicle, Money and their responsibilities using a simple domain scenario.	3	Talk	Think / Pair / Share
25	Identify relationships among Booking, Vehicle, Spot and Money. Draw a simple domain-object map.	3	Chalk	Problem-Based Learning
25	Discuss object guarantees and write sentence invariants for Booking, Vehicle and Spot.	3	Chalk	Case Study Based Learning
30	Prepare a domain diagram and invariant catalogue containing at least three rules for selected objects.	3	LTC	Problem-Based Learning
10	Present the domain map and explain the written invariants.	3	LTC	Peer Review

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce the three categories of invariants with Booking and ParkingSpot examples.	3	Talk	Quiz/Test Questions
25	Analyze single-field invariants such as positive amount and valid ID, and multi-field invariants involving related values.	3	Chalk	Problem-Based Learning
25	Explain temporal invariants using booking status transitions and parking-slot availability over time.	3	Chalk	Case Study Based Learning
30	Classify given rules and create one example for each invariant category in a worksheet.	3	LTC	Problem-Based Learning
10	Review classifications and correct mistakes.	3	LTC	Peer Review

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce object references, mutable objects and collections using a simple Java example.	3	Talk	Quiz/Test Questions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Explain public setters, escaping references and mutable elements inside collections.	3	Chalk	Problem-Based Learning
25	Analyze code examples in which an external object changes internal Booking or ParkingSpot data.	3	Chalk	Case Study Based Learning
30	Prepare a code-review sheet identifying unsafe references and propose defensive copying or immutable alternatives.	3	LTC	Problem-Based Learning
10	Present one reference hazard and its corrected design.	3	LTC	Peer Review

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce the problems of using ordinary String values for different identifiers.	3	Talk	Think / Pair / Share
25	Explain TicketId, SpotId, VehicleId and BookingId with simple examples.	3	Chalk	Problem-Based Learning
25	Analyze invalid states and explain how constructors, types and validation can prevent them.	3	Chalk	Case Study Based Learning
30	Design typed ID classes or records and prepare examples showing how the design prevents incorrect identifier usage.	3	LTC	Problem-Based Learning
10	Present the typed ID design and explain which invalid states are prevented.	3	LTC	Peer Review

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Review preconditions, postconditions and invariants using a booking allocation example.	4	Talk	Quiz/Test Questions
25	Analyze the SpotAllocator operation and identify caller responsibilities and expected results.	4	Chalk	Case Study Based Learning
25	Write preconditions and postconditions for allocateSpot and releaseSpot operations.	4	Chalk	Problem-Based

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				Learning
30	Prepare a contract sheet with valid inputs, expected results and failure conditions for domain services.	4	LTC	Problem-Based Learning
10	Review the contract sheet and check completeness.	4	LTC	Peer Review

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce base classes, subclasses and method overriding using a simple Java example.	4	Talk	Quiz/Test Questions
25	Explain Liskov Substitution Principle and the direction of precondition and postcondition changes.	4	Chalk	Problem-Based Learning
25	Analyze examples of stronger preconditions and weaker postconditions in overriding methods.	4	Chalk	Case Study Based Learning
30	Prepare a violation worksheet showing the caller expectation, incorrect override and corrected alternative.	4	LTC	Problem-Based Learning
10	Present violation and explain the corrected design	4	LTC	Peer Review

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce interfaces and explain why a single large interface may be difficult to maintain.	4	Talk	Think / Pair / Share
25	Explain role interfaces and Interface Segregation using Waypoint examples.	4	Chalk	Problem-Based Learning
25	Design common contract-test cases that each SpotAllocator implementation must satisfy.	4	Chalk	Case Study Based Learning
30	Prepare a reusable contract-test checklist and apply it to two alternative implementations.	4	LTC	Problem-Based Learning
10	Review the test suite and identify missing cases.	4	LTC	Peer Review

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce Java compiler errors and common mistakes in interface implementation and method overriding.	4	Talk	Quiz/Test Questions
25	Inspect error messages related to incorrect method signatures, missing implementations and invalid overrides.	4	Chalk	Problem-Based Learning
25	Analyze how incorrect implementations violate documented method contracts or caller expectations.	4	Chalk	Case Study Based Learning
30	Diagnose supplied Java code, correct the errors and rerun the shared contract tests.	4	LTC	Problem-Based Learning
10	Submit corrected code and explain the cause of one error.	4	LTC	Peer Review

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce coupling, cohesion and the meaning of change blast radius.	5	Talk	Quiz/Test Questions
25	Explain different dependencies among classes and identify responsibilities in a sample design.	5	Chalk	Problem-Based Learning
25	Prepare a simple cohesion matrix and trace the dependencies affected by a change request.	5	Chalk	Case Study Based Learning
30	Calculate affected-component counts and compare the change impact of two designs.	5	LTC	Problem-Based Learning
10	Recommend the design with lower change impact and justify the decision.	5	LTC	Peer Review

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce infrastructure dependencies such as external services and storage components.	5	Talk	Quiz/Test Questions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Draw a dependency map showing service classes and infrastructure dependencies.	5	Chalk	Problem-Based Learning
25	Explain the delete-the-infrastructure test and Law of Demeter using Java examples.	5	Chalk	Case Study Based Learning
30	Evaluate a tightly coupled service and redesign it using interfaces, constructor injection and limited object navigation.	5	LTC	Problem-Based Learning
10	Present the revised dependency diagram and explain the improvements.	5	LTC	Peer Review

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce refactoring and characterization tests using an existing Java class.	5	Talk	Quiz/Test Questions
25	Identify duplicated code, God Class responsibilities and tightly coupled methods.	5	Chalk	Problem-Based Learning
25	Prepare a refactoring plan and identify test cases that capture existing behaviour.	5	Chalk	Case Study Based Learning
30	Refactor the selected class in small steps and run characterization tests after each change.	5	LTC	Problem-Based Learning
10	Compare the original and refactored designs and review test results.	5	LTC	Peer Review

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce functional interfaces, lambda expressions and Java annotations.	5	Talk	Quiz/Test Questions
25	Analyze simple lambda expressions and compare them with anonymous-class implementations.	5	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Explain annotations such as @Override and introduce annotation processing and build-time validation.	5	Chalk	Case Study Based Learning
30	Improve a small Java example using lambdas and annotations; identify suitable compiler or build checks.	5	LTC	Problem-Based Learning
10	Present the code and explain why the selected Java features improve clarity or validation.	5	LTC	Peer Review

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce expected failures, programmer errors and transient infrastructure failures.	6	Talk	Quiz/Test Questions
25	Classify common Waypoint failures and discuss suitable responses for each category.	6	Chalk	Problem-Based Learning
25	Explain sealed failure types and Result-style success/ failure representations using simple Java examples.	6	Chalk	Case Study Based Learning
30	Design a sealed failure hierarchy and a Result-style flow for a booking operation.	6	LTC	Problem-Based Learning
10	Present the failure classification and demonstrate the expected control flow.	6	LTC	Peer Review

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce retry mechanisms and explain why repeatedly calling a failed service can create problems.	6	Talk	Quiz/Test Questions
25	Explain retry limits and exponential backoff using numerical examples.	6	Chalk	Problem-Based Learning
25	Analyze jitter, retry budgets and conditions under which a retry should stop.	6	Chalk	Case Study Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Prepare a retry policy for a Waypoint booking service with retry count, delay intervals and retry budget.	6	LTC	Problem-Based Learning
10	Present and review the retry policy.	6	LTC	Peer Review

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce multi-step booking and payment operations.	6	Talk	Quiz/Test Questions
25	Explain transactions and compensation using booking and payment examples.	6	Chalk	Problem-Based Learning
25	Identify successful and failed steps in a workflow and determine suitable compensation actions.	6	Chalk	Case Study Based Learning
30	Draw a saga workflow showing booking, payment, failure and compensation steps.	6	LTC	Problem-Based Learning
10	Present the recovery sequence and explain the compensation actions.	6	LTC	Peer Review

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce failure injection and duplicate-charge problems in a booking/payment workflow.	6	Talk	Quiz/Test Questions
25	Identify possible failure points such as timeout, payment failure and notification failure.	6	Chalk	Problem-Based Learning
25	Explain simple chaos-injection scenarios and idempotent consumer behaviour.	6	Chalk	Case Study Based Learning
30	Simulate repeated requests and failures using a small Java example or workflow model. Verify that duplicate charges are prevented.	6	LTC	Problem-Based Learning
10	Record expected and observed outcomes and present the results.	6	LTC	Peer Review

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce shared mutable state and explain why multiple threads accessing the same object can cause problems.	4	Talk	Quiz/Test Questions
25	Inspect Java classes and identify mutable fields and shared references.	4	Chalk	Problem-Based Learning
25	Classify fields as immutable, thread-confined or guarded using a thread-safety worksheet.	4	Chalk	Case Study Based Learning
30	Perform a mutable-field audit on a Waypoint class and recommend appropriate protection strategies.	4	LTC	Problem-Based Learning
10	Present the audit results and explain one concurrency risk.	4	LTC	Peer Review

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce multiple threads competing for a shared parking spot.	4	Talk	Quiz/Test Questions
25	Explain synchronized methods, locks and atomic variables using simple Java examples.	4	Chalk	Problem-Based Learning
25	Analyze executor-based tasks and CompletableFuture. Discuss contention and deadlock risks.	4	Chalk	Case Study Based Learning
30	Compare alternative concurrency approaches for a Waypoint operation and justify the selected mechanism.	4	LTC	Problem-Based Learning
10	Present the selected approach and explain its advantages and limitations.	4	LTC	Peer Review

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce visibility problems using a shared flag between two Java threads.	4	Talk	Quiz/Test Questions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Explain Java Memory Model and happens-before relationships using a simple thread timeline.	4	Chalk	Problem-Based Learning
25	Analyze volatile variables, synchronized blocks and safe publication.	4	Chalk	Case Study Based Learning
30	Trace a two-thread Java example and identify whether the required visibility guarantees are satisfied.	4	LTC	Problem-Based Learning
10	Present the thread timeline and explain the corrected code.	4	LTC	Peer Review

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce concurrent transactions using a Waypoint booking scenario.	4	Talk	Quiz/Test Questions
25	Explain transaction boundaries, aggregate consistency and basic isolation concepts.	4	Chalk	Problem-Based Learning
25	Analyze isolation anomalies and trace a simple write-skew example with two transactions.	4	Chalk	Case Study Based Learning
30	Prepare a transaction timeline showing how write skew occurs and recommend a safeguard such as stronger isolation or suitable locking/version checks.	4	LTC	Problem-Based Learning
10	Present the timeline and justify the selected consistency safeguard.	4	LTC	Peer Review

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce bounded contexts using Booking, Allocation and Payment examples.	6	Talk	Quiz/Test Questions
25	Identify different models and responsibilities across bounded contexts.	6	Chalk	Problem-Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	Explain anti-corruption layers and how they translate external data into the internal domain model.	6	Chalk	Case Study Based Learning
30	Draw a context map and design an anti-corruption layer for an external payment or allocation service.	6	LTC	Problem-Based Learning
10	Present the context map and explain how the boundary reduces coupling.	6	LTC	Peer Review

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce the Waypoint growth scenario and explain the meaning of 50× load.	6	Talk	Quiz/Test Questions
25	Explain request rates, users, workload assumptions and basic capacity calculations.	6	Chalk	Problem-Based Learning
25	Analyze caching and sharding approaches and discuss their benefits and trade-offs.	6	Chalk	Case Study Based Learning
30	Calculate estimated request load and prepare a scaling plan using suitable caching and sharding options.	6	LTC	Problem-Based Learning
10	Present the capacity worksheet and explain the assumptions used.	6	LTC	Peer Review

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce ADRs and explain why architecture decisions must be documented.	6	Talk	Quiz/Test Questions
25	Explain the structure of an ADR: context, decision, alternatives and consequences.	6	Chalk	Problem-Based Learning
25	Compare alternative architectural choices and identify their trade-offs and disadvantages.	6	Chalk	Case Study Based Learning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Prepare an ADR for a Waypoint design decision, including at least two alternatives and one rejected option.	6	LTC	Problem-Based Learning
10	Present the ADR and justify the final decision.	6	LTC	Peer Review

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduce the HLD dossier and review the required architecture evidence.	6	Talk	Quiz/Test Questions
25	Prepare UML class, sequence and state diagrams for Waypoint. C	6	Chalk	Problem-Based Learning
25	Organize ADRs, architecture tests, capacity estimates and relevant evidence.	6	Chalk	Case Study Based Learning
30	Assemble the HLD evidence pack and conduct a mock architecture defence covering design decisions, security, cost and SLOs.	6	LTC	Problem-Based Learning
10	Present final observations, receive feedback and record improvements.	6	LTC	Peer Review

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	H-S1,H-S2,H-S3	H-S1,H-S2,H-S3	---	---	--	---	--	---	---
	Tutorial	--	--	---	---	--	---	--	---	---
	Lab	--	--	---	---	--	---	--	---	---
	Skilling	--	--	---	---	H-S4,H-S4,H-S5,H-S5,H-S6,H-	---	H-S4,H-S4,H-S5,H-S5,H-S6,H-	---	---

	Hour	1	2	3	4	5	6	7	8	9
						S6		S6		
Tue	Theory	H-S4,H-S5,H-S6	H-S4,H-S5,H-S6	--	--	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	---	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	---	---
	Skilling	--	--	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	--	---	--	---	---
Wed	Theory	H-S7,H-S8,H-S9	H-S7,H-S8,H-S9	--	--	---	---	---	---	---
	Tutorial	--	--	--	--	---	---	---	---	---
	Lab	--	--	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	---	---	---	---	---
	Skilling	--	--	--	--	---	---	---	---	---
Thu	Theory	--	--	---	---	--	---	--	---	---
	Tutorial	--	--	---	---	--	---	--	---	---
	Lab	--	--	---	---	--	---	--	---	---
	Skilling	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	---	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	---
Fri	Theory	--	--	--	--	---	---	---	---	---
	Tutorial	--	--	--	--	---	---	---	---	---
	Lab	--	--	--	--	---	---	---	---	---
	Skilling	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	---	---	---	---
Sat	Theory	---	---	---	---	--	---	--	---	---
	Tutorial	---	---	---	---	--	---	--	---	---
	Lab	---	---	---	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9	---	---
	Skilling	---	---	---	---	--	---	--	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--

	Hour	1	2	3	4	5	6	7	8	9
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCS
1	Investigate the principles of domain-driven object modelling and explore how domain objects, object identity, value objects and invariants help maintain data consistency in enterprise applications.	CO1	Case Study Based Learning	1. Effective Java Joshua Bloch Addison-Wesley 2018 (3rd Ed.) 2. A Philosophy of Software Design John Ousterhout Yaknyam Press 2021 (2nd Ed.) 3. Domain-Driven Design: Tackling Complexity in the Heart of Software Eric Evans Addison-Wesley 2003
2	Examine immutability, defensive copying, typed identifiers and encapsulation techniques, and investigate their role in preventing invalid states in object-oriented applications.	CO1	Case Study Based Learning	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153
3	Investigate Design by Contract principles, including preconditions, postconditions and invariants, and analyze how they improve the reliability and correctness of object-oriented systems.	CO2	Case Study Based Learning	Advanced Object-Oriented Programming is the Regular-track sister of PSPJ's object model and the PBL design studio for CSE: a twelve-week, project-based course in

S.no	Topics	CO	ALM	References/MOOCS
				software design — LLD and HLD — rather than in syntax students already have. “Design is what you can change on a Tuesday”: the course teaches the decisions that determine whether next quarter’s requirement costs two files or two weeks, and the habit of pricing every decision aloud. INSIDE-OUT, in five movements: Modelling (objects that hold their shape, contracts before code, the cost of change), Principles (SOLID applied, the type system as a design tool, failure as a first-class design concern), Patterns (creational, structural and behavioural, each selected by problem shape and costed), Runtime (concurrency as design, persistence and aggregates), and Architecture & Capstone (bounded modules, capacity arithmetic, shipping it). Every week is a full build on one repository — Waypoint, an urban mobility and payments platform — with four bespoke interactive instruments, a debugging lab of four real failures, starter code with failing tests, and a gated reference; six studio builds (SeatLock, LedgerPay, FleetMatch, GateKeeper, HeraldQ,

S.no	Topics	CO	ALM	References/MOOCS
				MatchBook) are the transfer test, and each is a famous interview problem in domain clothes. Language topics — generics, lambdas and streams, reflection, serialization, JVM behaviour, C++ and Python equivalents — are taught where a design problem needs them and collected in the Language Annex, not as chapters. Honest scope: it does not teach DSA, a framework, front-end, DevOps tooling or behavioural rounds. Pre-requisite: PSPJ — Java 17+, classes and interfaces, generics as a consumer, collections, exceptions; plus Gradle/Maven, JUnit 5, Git, and basic SQL from Week 11; concurrency is not assumed. Enables: SDSR — System Design for Scalability, and every backend, platform and LLD/HLD interview-loop course downstream. Placement corridor: the claim the course makes is precise — a student who completes it can design, defend and extend a non-trivial system and perform in the LLD and HLD rounds of a product-company interview loop; ₹8–22 LPA backend and platform-engineering roles at India product companies, with the DSA leg covered by the

S.no	Topics	CO	ALM	References/MOOCS
				DSA plan and the behavioural leg by Professional Presence.
4	Explore the Liskov Substitution Principle (LSP), interface-based programming and polymorphism, and investigate how substitutability violations affect software maintainability.	CO2	Case Study Based Learning	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153
5	Examine the SOLID principles, coupling and cohesion, and investigate how dependency inversion and dependency injection improve modularity and reduce change impact in enterprise applications.	CO3	Case Study Based Learning	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153
6	Investigate commonly used object-oriented design patterns and analyze the applicability of Strategy, Factory, Builder and Observer patterns in developing maintainable software systems.	CO3	Case Study Based Learning	1. Implementing Domain-Driven Design Vaughn Vernon Addison-Wesley 2013 2. Design Patterns: Elements of Reusable Object-Oriented Software Erich Gamma, Richard Helm, Ralph Johnson & John Vlissides Addison-Wesley 1994 3. Refactoring: Improving the Design of Existing

S.no	Topics	CO	ALM	References/MOOCS
				Code Martin Fowler Addison-Wesley 2018 (2nd Ed.)
7	Explore exception handling, timeout management, retry mechanisms and idempotency, and investigate how these techniques help build reliable distributed application workflows.	CO4	Case Study Based Learning	1. Implementing Domain-Driven Design Vaughn Vernon Addison-Wesley 2013 2. Design Patterns: Elements of Reusable Object-Oriented Software Erich Gamma, Richard Helm, Ralph Johnson & John Vlissides Addison-Wesley 1994 3. Refactoring: Improving the Design of Existing Code Martin Fowler Addison-Wesley 2018 (2nd Ed.)
8	Examine Circuit Breaker, Compensating Transactions and Transactional Outbox patterns, and analyze their significance in handling failures and ensuring reliable event processing.	CO4	Case Study Based Learning	1. Effective Java Joshua Bloch Addison-Wesley 2018 (3rd Ed.) 2. A Philosophy of Software Design John Ousterhout Yaknyam Press 2021 (2nd Ed.) 3. Domain-Driven Design: Tackling Complexity in the Heart of Software Eric Evans Addison-Wesley 2003
9	Investigate concurrency concepts, shared mutable state, synchronization and thread safety, and explore their role in developing reliable multithreaded Java applications.	CO5	Case Study Based Learning	1. Effective Java Joshua Bloch Addison-Wesley 2018 (3rd Ed.) 2. A Philosophy of Software Design John Ousterhout Yaknyam Press 2021 (2nd Ed.) 3. Domain-Driven Design: Tackling Complexity in the Heart of Software Eric Evans Addison-Wesley 2003
10	Examine database transaction isolation levels, race conditions and write skew, and investigate how synchronization and transaction management help maintain consistency in concurrent applications.	CO5	Case Study Based Learning	1. Effective Java Joshua Bloch Addison-Wesley 2018 (3rd Ed.) 2. A Philosophy of Software Design John Ousterhout

S.no	Topics	CO	ALM	References/MOOCS
				Yaknyam Press 2021 (2nd Ed.) 3. Domain-Driven Design: Tackling Complexity in the Heart of Software Eric Evans Addison-Wesley 2003
11	Explore domain-driven design concepts such as bounded contexts, module boundaries and scalable architecture, and investigate their applications in designing enterprise-level software systems.	CO6	Case Study Based Learning	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153
12	Investigate UML modelling, Architecture Decision Records (ADRs) and capacity estimation techniques, and analyze their importance in documenting and evaluating scalable software architectures.	CO6	Case Study Based Learning	1. https://www.coursera.org/specializations/software-design-architecture 2. https://www.coursera.org/learn/object-oriented-design 3. https://www.coursera.org/learn/design-patterns 4. https://www.coursera.org/learn/software-architecture 5. https://nptel.ac.in/courses/106105153

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCS

EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/ Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Term End Project Defence & Demonstration	Weightage	16		180	3	2.5	2.5	2.5	2.5	3
		Max Marks	60			10	10	10	10	10	10
	End Semester Exam	Weightage	24		180	2	2	2	6	6	6
		Max Marks	90			5	5	5	25	25	25
In Semester Formative Evaluation Total= 36 %	Global Challenges - Participation	Weightage	12		90	2	2	2	2	2	2
		Max Marks	60			10	10	10	10	10	10
	Active Learning Engagement (In-Class)	Weightage	8		90	1	1	1	2	2	1
		Max Marks	60			10	10	10	10	10	10
	Practical Continuous Evaluation	Weightage	8		90	1	1.5	1.5	1.5	1.5	1
		Max Marks	60			10	10	10	10	10	10
	Project Continuous Evaluation	Weightage	8		90	1.5	1	1	1.5	1.5	1.5
		Max Marks	60			10	10	10	10	10	10
In Semester Summative Evaluation Total= 24 %	In-Term Practical Examination	Weightage	6		100	1	1	1	1	1	1
		Max Marks	50			5	5	10	10	10	10
	Mid-Term Examination (Descriptive)	Weightage	12		90	4	4	4			
		Max Marks	45			15	15	15			
	MOOCs Evaluation (Certificate + Test)	Weightage	6		90	1	1	1	1	1	1
		Max Marks	60			10	10	10	10	10	10

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Yamini Jonnala	L	8-MA	-	-	-	-
Yamini Jonnala	P	8-A,5-B	-	-	-	-
Yamini Jonnala	S	8-A,3-B,5-B	-	-	-	-
DHULIPALLA KRISHNA	L	4-MA	-	-	-	-
DHULIPALLA KRISHNA	P	9-B,4-A,1-B	-	-	-	-
DHULIPALLA KRISHNA	S	9-B,1-B,4-A	-	-	-	-
Karumuri Murthy	L	6-MA	-	-	-	-
Karumuri Murthy	P	2-B,6-A,7-B	-	-	-	-
Karumuri Murthy	S	6-A,2-B	-	-	-	-
Ramatenki Kumar	L	1-MA	-	-	-	-
Ramatenki Kumar	P	1-A,4-B	-	-	-	-
Ramatenki Kumar	S	4-B,7-B,1-A	-	-	-	-
Pratima Depa	L	3-MA	-	-	-	-
Pratima Depa	P	8-B,6-B,3-A	-	-	-	-
Pratima Depa	S	8-B,3-A	-	-	-	-
Lakshmi Karthek Vankamamidi	L	2-MA,7-MA	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Lakshmi Karthek Vankamamidi	P	2-A,7-A	-	-	-	-
Lakshmi Karthek Vankamamidi	S	5-B,6-B,7-A,2-A	-	-	-	-
RUDRA kRISHNASRIJA	L	9-MA,5-MA	-	-	-	-
RUDRA kRISHNASRIJA	P	3-B,9-A,5-A	-	-	-	-
RUDRA kRISHNASRIJA	S	3-B,5-A,9-A	-	-	-	-
Sama Mineesha	S	6-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(Karumuri Sri Rama Murthy)(9870)(CSE-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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